



Where do Gradient Methods Really Happen? Maybe Not Where We Expected

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Intro

Modern neural networks are trained in extremely high-dimensional parameter spaces, yet their dynamics often look partly low-dimensional. Recent work observes that gradients may quickly align with a dominant Hessian subspace after only a few training steps. Our draft revisits this observation: strong alignment with a dominant subspace does not necessarily mean that useful learning actually happens inside it. The goal is to separate descriptive geometry from causal optimization mechanisms across architectures, optimizers, and alternative choices of training subspaces.

Problem

Let u_t be the update produced by an optimizer from the current gradient and optimizer state. For a subspace U_t , we decompose this update into projected and complementary parts:

$$u_t = P_{U_t} u_t + (I - P_{U_t}) u_t.$$

We compare three trajectories from the same initialization: full updates, updates restricted to U_t , and updates restricted to the orthogonal complement. The central question is causal: does a high-alignment subspace preserve loss decrease and validation quality, or does the complementary “bulk” still carry the useful progress?

Projectors

The poster will compare several ways to choose U_t :

- top Hessian eigenspaces, following the dominant-subspace hypothesis;
- gradient covariance subspaces and random baselines;
- optimizer-induced subspaces from momentum or adaptive learning rates;
- learnable orthonormal bases on the Stiefel manifold.

In the current implementation, the main working projectors are top- k Hessian eigenspaces and random subspaces, with a learnable projector smoke test already indicating that non-Hessian subspaces can be meaningful.

Optimizers

We treat the optimizer as part of the geometry, not as a detail outside the projection experiment. The update u_t may include momentum, adaptive scaling, matrix normalization, or finite-difference estimates. The experimental code is organized around a shared runner for:

- SGD and SGD with momentum;
- Adam and AdamW;
- Muon and related matrix-aware updates;
- forward-gradient and zeroth-order methods such as MeZO/SubZero.

This lets us test whether the dominant-subspace failure is specific to SGD-like training or persists under richer optimizer-induced geometries.

Experiments

The current draft contains a unified experiment framework and initial replications on:

- toy binary classification;
- MLP on MNIST-5k and CNN on CIFAR10-5k;
- a small Transformer on SST2-1k;
- additional architectures such as VGG11 and ResNet8.

Preliminary results reproduce the key phenomenon: top Hessian projection can stop or degrade training, while the complementary subspace may continue reducing loss. Early Muon and momentum-induced subspace experiments suggest that optimizer-specific subspaces may behave differently and may be relevant for validation behavior, not only training loss.

Conclusions

The working conclusion is that dominant Hessian alignment is not a sufficient explanation of learning dynamics. A simple low-rank quadratic view already explains why projection onto sharp dominant directions may stop once those coordinates are optimized, even when the full gradient is not zero. The final version will compare optimizer-projector pairs and identify when useful progress lives in dominant, bulk, random, learned, or optimizer-induced subspaces.

Main message. Low-dimensional structure is real, but the useful learning signal may live outside the dominant Hessian subspace. Alignment must be tested against projected training quality.

Metrics

We measure subspace alignment by

$$\chi_U(v) = \frac{\|P_U v\|}{\|v\|}.$$

Following the reference setup, an exponential moving average of alignment can be used as a switching criterion; however, the draft emphasizes that alignment alone is not enough. We therefore track training loss, validation loss, accuracy, update norms, and a projected descent ratio: how much predicted one-step loss decrease remains after projection. These metrics distinguish geometric overlap from actual optimization and generalization behavior.

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